

Title:

Green Hydrogen from UK Offshore Wind: Comparison of Water Electrolysis Technologies with Water Treatment, Hydrogen Storage, and Ammonia Transportation at the Arven Site, Shetland, Scotland



Research Dissertation Proposal Report

Submitted by:

Syed Yousaf Munib (6933767)

Project Supervisor :

Dr Bahman Amini Horri

Submission Date: 10 June 2026

Declaration of originality

"I confirm that the submitted work is my own work. No element has been previously submitted for assessment, or where it has, it has been correctly referenced. I have clearly identified and fully acknowledged all material that is entitled to be attributed to others (whether published or unpublished) using the referencing system set out in the programme handbook.

I agree that the University may submit my work to means of checking this, such as the plagiarism detection service Turnitin® UK and the Turnitin® Authorship Investigate service. I confirm that I understand that assessed work that has been shown to have been plagiarised will be penalised.

If in completing this work I have been assisted with its presentation by another person, I will state their name and contact details of the assistant in the 'Comments' text box below. In addition, if requested, I agree to submit the draft material that was completed solely by me prior to its presentational improvement."

Syed Yousaf Munib



Table of Contents

1. Introduction	7
1.1 Problem Statement	9
1.2 Research Aim and Objectives	10
1.3 Research Motivation	11
2. Literature Review	12
2.1 Comparative Assessment of Water Electrolysis Technologies	12
2.1.1 Alkaline Water Electrolysis (AWE)	12
2.1.2 Proton Exchange Membrane (PEM) Water Electrolysis	13
2.1.3 Solid Oxide Water Electrolysis (SOEC)	14
2.2 Comparative Synthesis and Conflicting Findings	17
2.3 Research Gap	18
3. Methodology	19
3.1 Research Design and System Boundary	19
3.2 Technology Selection	21
3.3 Aspen Plus Process Modelling	21
3.4 Mass and Energy Analysis	22
3.5 Techno-Economic Analysis (TEA)	24
3.5.1 CAPEX and OPEX Modelling	24
3.5.2 Financial Metrics	25
4. Expected Contributions	27
5. Ethical Considerations	28
Timeline	29
References	30
Appendices	34

List of Figures

Figure 1. Comparison between the full planned Arven offshore wind capacity and the 50 MW Aspen Plus design basis	8
Figure 2. Schematic illustration of the alkaline water electrolysis working principle, showing hydroxyl-ion (OH^-) transport across the porous diaphragm.....	13
Figure 3. Schematic view of the PEM water electrolysis working principle, showing proton (H^+) transport through the solid polymer membrane.....	14
Figure 4. Schematic view of the solid oxide water electrolysis (SOEC) working principle, showing oxide-ion (O^{2-}) transport through the dense ceramic electrolyte at high temperature	15
Figure 5. Conceptual overview of research objectives and methodology focus	20
Figure 6. Methodology flowchart for the 50 MW offshore power-to-ammonia model.....	27
Figure 7. Objective based Gantt chart showing the planned dissertation timeline from February to September 2026.....	29

List of Tables

Table 1. Simple comparison of AWE, PEM and SOEC for offshore wind-to-hydrogen production.....	16
Table 2. Link between research problems, objectives and methodology.....	20
Table 3. Main Aspen Plus design assumptions.	22

List of Symbols

Symbol	Description	Unit
$P_{\text{installed}}$	Installed offshore wind capacity	MW
$P_{\text{available}}$	Effective power available for PEM electrolysis	MW
P_{net}	Net power after parasitic loads	MW
$P_{\text{parasitic}}$	Total parasitic power demand	MW
CF	Capacity factor	–
SEC_{PEM}	Specific energy consumption of PEM electrolyser	kWh/kg H_2
$\dot{m}_{H_2}$	Hydrogen mass flow rate	kg/h
$NH_{3,\text{daily}}$	Daily ammonia production	t/day
$NH_{3,\text{annual}}$	Annual ammonia production	t/year
$CAPEX_{\text{total}}$	Total capital expenditure	USD
$CAPEX_{\text{ann.}}$	Annualised capital expenditure	USD/yr
$OPEX_{\text{ann.}}$	Annual operating expenditure	USD/yr
f_{offshore}	Offshore location cost factor	–
CRF	Capital recovery factor	–
i	Discount rate	–
n	Project lifetime	years
$LCOA$	Levelised cost of ammonia	USD/t
P_{NH_3}	Ammonia selling price	USD/t
SPP	Simple payback period	years
η_{PEM}	PEM electrolyser efficiency	–
ΔG	Gibbs free energy change of water splitting	kJ/mol

1. Introduction

Green hydrogen is becoming an important option for reducing carbon emissions in sectors where direct electrification is difficult. In the UK, this is linked to the national Net Zero target by 2050 and the aim to develop up to 10 GW of low carbon hydrogen production capacity by 2030 (HM Government, 2021; DESNZ, 2024). Its wider use also depends on how it is stored, transported and connected to renewable energy systems.

The Arven Offshore Wind Farm, east of the Shetland Islands, is planned as a 2.3 GW floating offshore wind development, which makes it a suitable case study for offshore hydrogen and ammonia production (Ocean Winds, 2025). However, offshore wind is variable, so the selected electrolyser must respond well to changing power output. At the same time, PEM electrolysis has limitations, including high cost, membrane degradation and the need for very pure water.

The water supply is a critical factor as seawater is not an ideal source for PEM electrolysis. The presence of salts and impurities in seawater can lead to damage of the membrane and decreased performance of the electrolysis process; therefore, desalination and purification will need to be incorporated into the overall system design. Furthermore, transporting hydrogen from remote offshore sites will be difficult due to the low volumetric energy density of compressed hydrogen. Ammonia offers a possible solution because it is easier to store and transport than gaseous hydrogen.

This proposal evaluates a 50 MW modular offshore power-to-ammonia system at the Arven site using Aspen Plus. The model links offshore wind power, water treatment, PEM electrolysis, hydrogen production, nitrogen supply and ammonia synthesis. The aim is not to design a full industrial plant, but to assess the technical and economic feasibility of a realistic modular system and identify the main limitations for future scale-up.

To keep the Aspen Plus simulation realistic and manageable, this proposal uses a reduced 50 MW modular offshore production unit rather than modelling the full Arven offshore wind farm. A 50 MW model is sufficient to assess the main mass balance, energy balance and techno-economic behaviour of an offshore power-to-ammonia system.

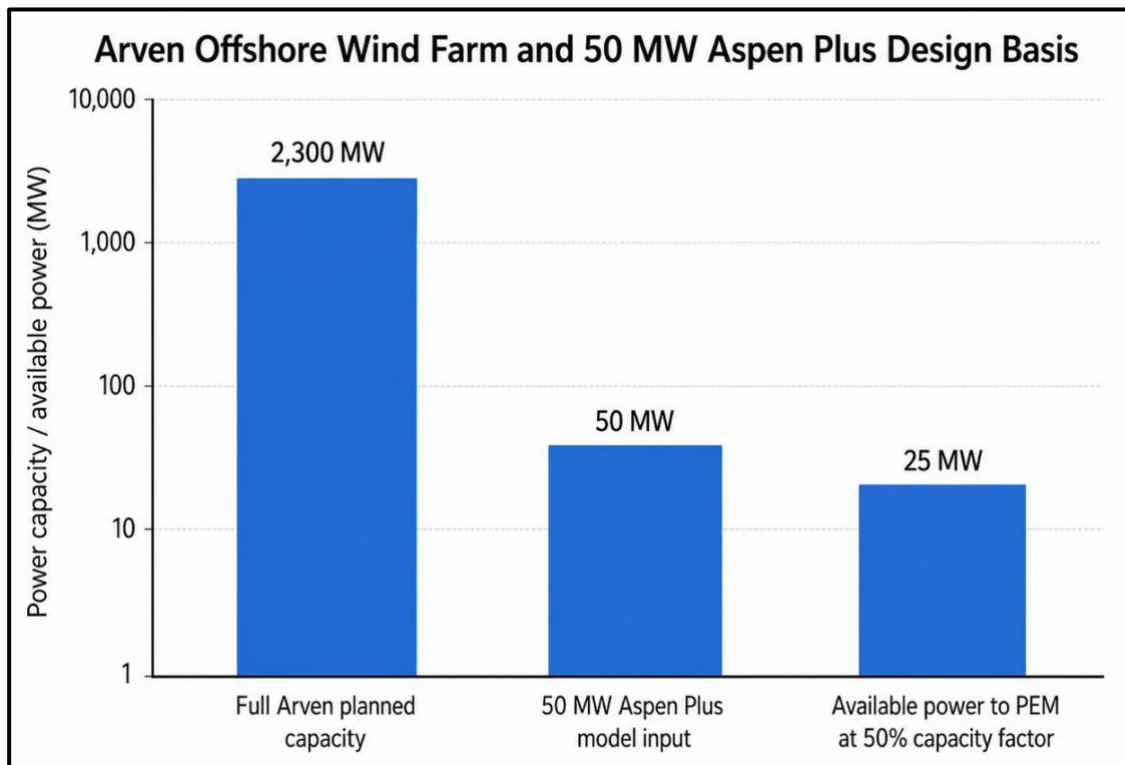


Figure 1. Comparison between the full planned Arven offshore wind capacity and the 50 MW Aspen Plus design basis (Appendix 1).

Figure 1 compares the full planned Arven generation capacity with the 50 MW Aspen Plus model input and the 25 MW effective available power after applying a 50% capacity factor. A logarithmic scale is used to make the smaller 50 MW and 25 MW values visible.

1.1 Problem Statement

The development of an offshore power-to-ammonia system at the Arven site in Shetland faces several technical and economic challenges. Although the site has strong offshore wind potential, the variable nature of wind power makes hydrogen production more complex. The selected electrolyser must be able to respond to changing power output without causing major losses in efficiency, durability or system performance. This creates a need to compare AWE, PEM and SOEC technologies and justify the most suitable option for offshore operation (Buttler and Spliethoff, 2018; Wei et al., 2019).

Many studies consider hydrogen production as a separate process, but offshore ammonia production also requires seawater treatment, PEM electrolysis, nitrogen supply, ammonia synthesis, storage and transport. If these stages are not assessed together, the total energy demand and system performance may be underestimated (Lee et al., 2021; IRENA and AEA, 2022).

The Arven site is located far from major demand centres, so transporting gaseous hydrogen from offshore Shetland to the UK mainland is difficult due to its low volumetric energy density. Converting hydrogen into ammonia may offer a more practical transport route, but this also adds extra energy demand, process complexity and safety considerations (IEA, 2021; IRENA and AEA, 2022).

The capital cost, operating cost, levelised cost of ammonia and payback period are not yet clear for a modular 50 MW offshore system (Schmidt et al., 2017). Therefore, this study addresses the problem by developing an Aspen Plus model and techno-economic assessment to evaluate whether offshore wind-based ammonia production at the Arven site is technically feasible and economically reasonable.

1.2 Research Aim and Objectives

The aim is to assess the technical and economic feasibility of producing green ammonia using offshore wind at the Arven site in Shetland. The performance of a complete production chain will be assessed including water treatment, PEM electrolysis, hydrogen production, nitrogen supply, ammonia synthesis, and ammonia distribution.

To achieve the research aim, the following technical objectives will be addressed:

1. **To compare** AWE, PEM and SOEC technologies and justify the selection of PEM electrolysis for variable offshore wind operation based on flexibility, efficiency, durability, cost and suitability for the Arven site.
2. **To develop** an integrated Aspen Plus model of a 50 MW offshore power-to-ammonia system, including desalinated water supply, PEM hydrogen production, nitrogen supply and ammonia synthesis.
3. **To quantify** the mass and energy balance of the system including available wind power, water demand, hydrogen production rate, nitrogen requirement, ammonia production rate and the main parasitic energy demand from purification, compression and liquefaction.
4. **To assess** the storage and transport pathway by evaluating how hydrogen buffering and conversion to ammonia affect system efficiency, process complexity, safety considerations and offshore logistics.
5. **To evaluate** the techno-economic feasibility of the proposed system by estimating CAPEX, OPEX, levelised cost of ammonia and simple payback period, testing the sensitivity of key variables such as capacity factor, electrolyser energy use, ammonia conversion and production rate.

1.3 Research Motivation

The motivation for this research comes from the UK need to reduce carbon emissions and develop secure low carbon energy systems. The UK has committed to achieving Net Zero by 2050, and low carbon hydrogen is expected to support sectors that are difficult to electrify directly such as heavy industry, shipping and long distance energy storage (HM Government, 2021; DESNZ, 2024). The Arven site in Shetland provides a relevant case study because it is linked to large-scale offshore wind development and is located in an area with strong renewable energy potential.

A key motivation for this project is the challenge of using offshore wind power effectively. Offshore wind can produce large amounts of renewable electricity but its output changes with wind conditions. If this electricity cannot be used stored or converted into another energy carrier part of its value may be lost. Green hydrogen offers one route for storing renewable electricity but transporting hydrogen from a remote offshore location is difficult because of its low volumetric energy density. Converting hydrogen into green ammonia may provide a more practical transport pathway because ammonia is easier to store and ship than gaseous hydrogen.

This research is also motivated by the need for a clearer technical and economic assessment of offshore green ammonia production. Many studies discuss green hydrogen and ammonia at a general level, but fewer studies connect offshore wind power, water treatment, PEM electrolysis, hydrogen production, nitrogen supply, ammonia synthesis and transport in one integrated model. By using Aspen Plus and a 50 MW modular design basis, this proposal aims to assess whether offshore wind based ammonia production at the Arven site is technically feasible and economically reasonable. The study also recognises that the model is simplified, so the results will be used as an early-stage assessment rather than a final industrial design.

2. Literature Review

2.1 Comparative Assessment of Water Electrolysis Technologies

Hydrogen can be produced through water electrolysis, where electricity is used to split water into hydrogen and oxygen. When powered by renewable electricity, this process can support the decarbonisation of sectors where direct electrification is difficult (Ibrahim, 2012). Thermodynamically the reaction demands a minimum cell voltage of 1.23 V at ambient conditions, although a practical voltage closer to 1.48 V is required once kinetic and ohmic losses are accounted for (Carmo et al., 2013). Electrolytic hydrogen still accounts for only a small fraction of global supply (Yu et al., 2018). Electrolyser families are conventionally distinguished by their electrolyte and charge carrier alkaline (AWE), proton exchange membrane (PEM) and solid oxide (SOEC) systems (Nikolaidis & Poullikkas, 2017). The comparative analysis below reviews independent evaluations of offshore systems based on their performance relative to four criteria: Dynamic Response, Efficiency, Durability and Price. Where independent studies disagree on a given aspect of a system performance, the nature of that disagreement will be described herein, since it is through analysis of these points of contention that the area of opportunity addressed by this report has been identified (Lee et al., 2021).

2.1.1 Alkaline Water Electrolysis (AWE)

Alkaline electrolysis is the most mature of the three technologies considered operated commercially at multi-megawatt scale since the first industrial plant of 1939 (Diogo et al., 2013). As Figure 2 shows, the cell relies on a concentrated 5 M KOH or NaOH solution in which hydroxyl ions (OH^-) carry charge from cathode to anode across a porous diaphragm (Shiva Kumar et al., 2017). Reviews broadly agree on its appeal inexpensive nickel catalysts, lifetimes near 90,000 hours and capital costs of roughly USD 500-1000/kW (Kuckshinrichs et al., 2017).

AWE flexibility remains contested: pressurised stacks may follow renewable loads, but liquid-electrolyte inertia and gas-management issues restrict rapid response. Key limits include low current density and gas crossover, reducing hydrogen purity (Brauns and Thomas, 2020; Buttler and Spliethoff, 2018; Grigoriev et al., 2020).

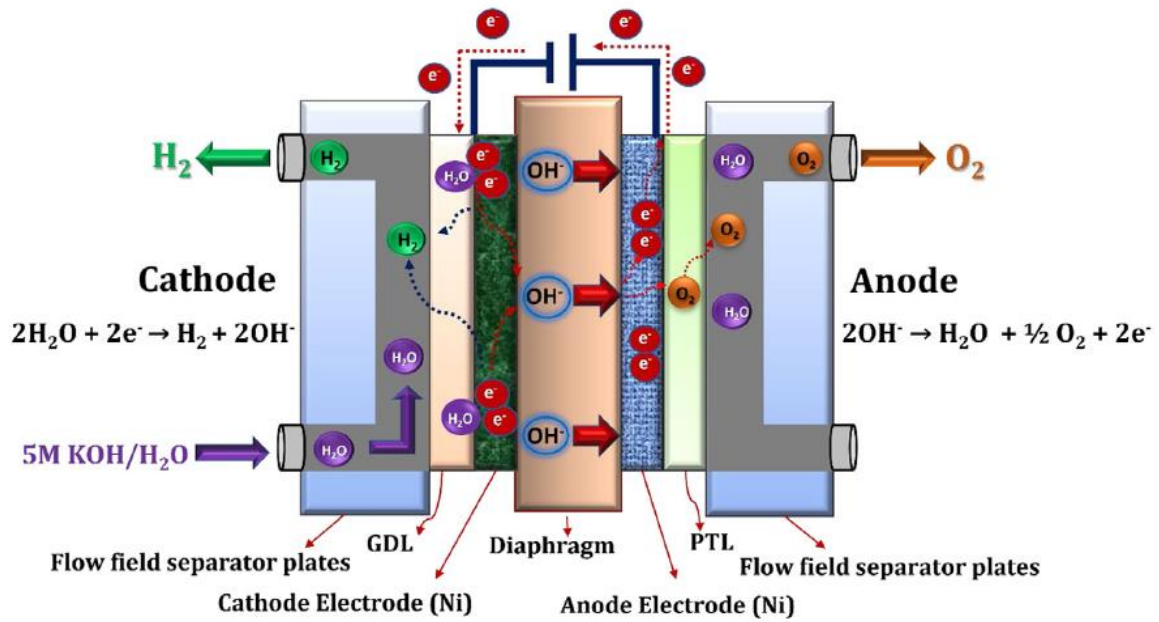


Figure 2. Schematic illustration of the alkaline water electrolysis working principle, showing hydroxyl-ion (OH^-) transport across the porous diaphragm (Shiva Kumar & Lim, 2022).

2.1.2 Proton Exchange Membrane (PEM) Water Electrolysis

PEM electrolysis addresses several AWE limitations by using a solid polymer membrane and deionised water to produce compact, high-purity hydrogen systems. As Figure 3 shows, protons migrate through a Nafion type membrane, enabling current densities of 1-2 A/cm² and hydrogen purity up to 99.999% (Carmo et al., 2013; Shiva Kumar and Lim, 2022). Its main advantage for offshore wind is rapid transient response, with intermittent operation showing limited immediate performance loss (Wei et al., 2019). However, variable loading may accelerate degradation, while costs remain strongly linked to titanium bipolar plates and noble metal catalysts (Schmidt et al., 2017; Yang et al., 2019; Min et al., 2021). Nevertheless, most PEM intermittency evidence is still based on controlled or laboratory power profiles, so its long-term durability under real offshore wind variability remains uncertain. Catalyst research illustrates the same bench versus stack gap seen for AWE: the ultrathin platinum nanowire electrode of Xie et al. (2021) achieved a notably low cell voltage of 1.643 V at 1 A/cm² with greatly reduced loading, but such results are reported at small scale

and rarely accompanied by the thousand hour durability data that would confirm field viability (Tajuddin et al., 2021).

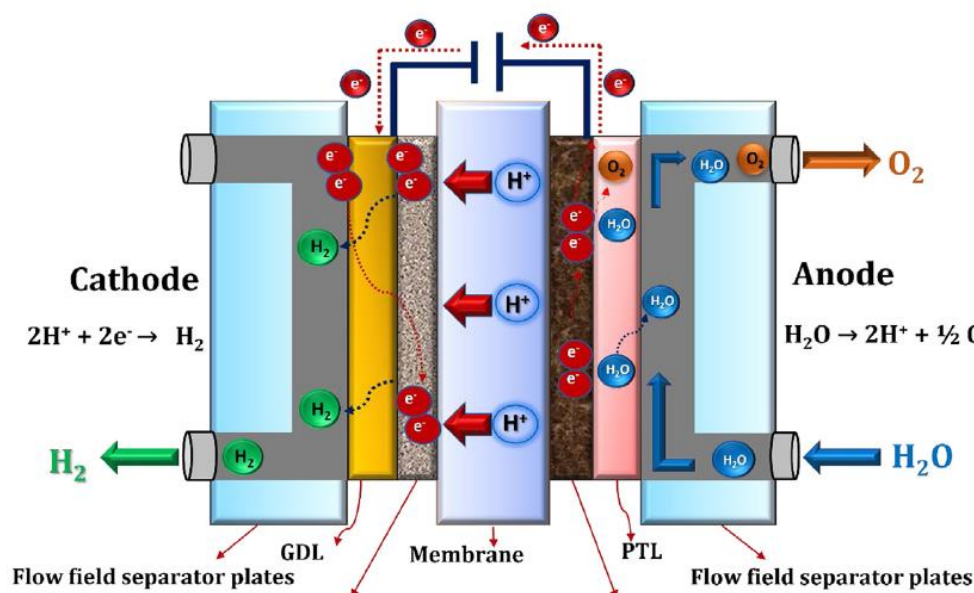


Figure 3. Schematic view of the PEM water electrolysis working principle, showing proton (H^+) transport through the solid polymer membrane (Shiva Kumar & Lim, 2022).

2.1.3 Solid Oxide Water Electrolysis (SOEC)

Solid oxide electrolysis processes steam at 500-850 °C, which sharply lowers the electrical input and lifts conversion efficiency (Buttler & Spliethoff, 2018). Figure 4 shows a dense yttria stabilised zirconia (YSZ) electrolyte conducting oxide ions (O^{2-}) between a Ni-YSZ fuel electrode and perovskite oxygen electrodes such as LSCF and LSM (Hauch et al., 2020). Proponents emphasise laboratory efficiencies approaching 89% and the ability to couple thermally with ammonia or methanol synthesis (Khan et al., 2018). The persistent counter argument concerns durability, and the studies disagree sharply on how severe the problem is. Tietz et al. (2013) document measurable electrode degradation after 9,000 hours, and Kukk et al. (2021) report a markedly higher degradation rate in electrolysis mode (16.3%) than in fuel cell mode for the same materials, attributing it to silicon and chromium impurities and electrode cracking. Against this, recent materials-focused studies suggest that SOEC degradation can be reduced through improved electrode and cell materials. However, these improvements are often demonstrated under controlled laboratory conditions,

while long-term stack operation under realistic cycling remains a major challenge (Wang et al., 2020; Nechache and Hody, 2021). Crucially for the present application, almost none of the SOEC durability literature tests the repeated cold-start thermal cycling that an offshore platform on intermittent wind would impose, which is the very condition under which the technology is most vulnerable (Nechache & Hody, 2021). Combined with the highest capital cost of the group at over USD 2000/kW, this leaves SOEC promising in principle but unproven for the duty cycle examined here.

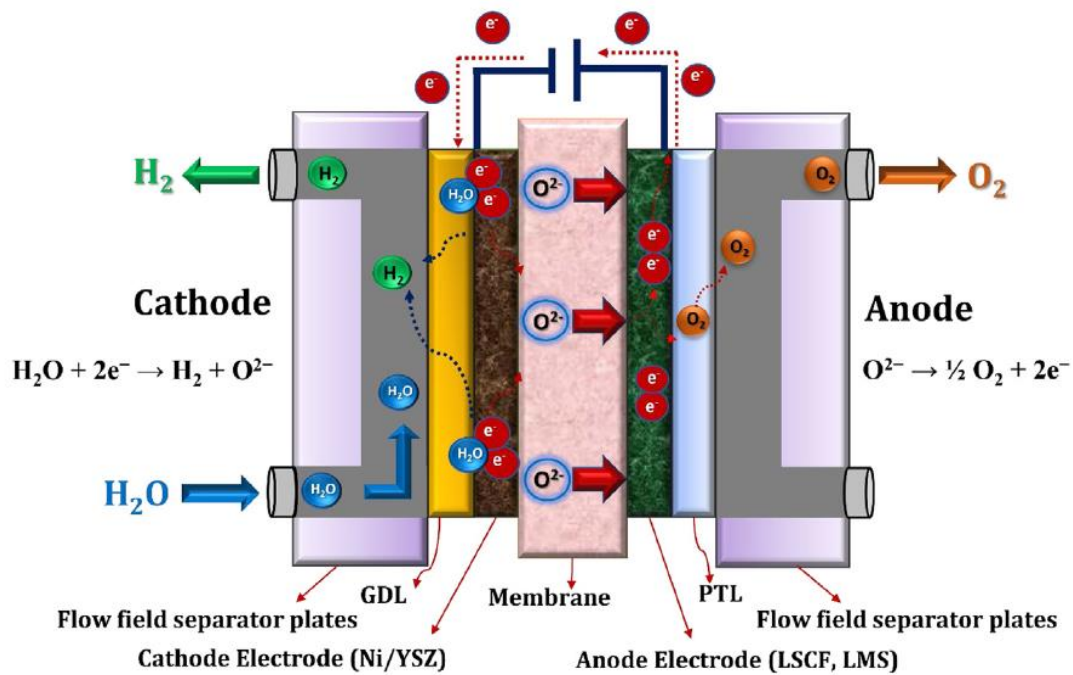


Figure 4. Schematic view of the solid oxide water electrolysis (SOEC) working principle, showing oxide-ion (O^{2-}) transport through the dense ceramic electrolyte at high temperature (Shiva Kumar & Lim, 2022).

The main differences between AWE, PEM and SOEC technologies using the key criteria relevant to offshore wind-to-hydrogen production.

Criteria	AWE	PEM	SOEC	Key sources
Technology maturity	Mature, commercial, proven	Commercial, developing	Less mature, research stage	(Diogo et al., 2013; Nikolaidis & Poullikkas, 2017; Kuckshinrichs et al., 2017)
Dynamic response	Slower response, less flexible	Fast response, good for wind	Limited by high temperature	(Buttler & Spliethoff, 2018; Carmo et al., 2013; Wei et al., 2019)
Efficiency	Moderate efficiency	Good efficiency, compact system	High efficiency potential	(Carmo et al., 2013; Buttler & Spliethoff, 2018; Khan et al., 2018)
Durability	Long lifetime under steady operation	Good lifetime, but degradation risk	Thermal cycling and degradation issues	(Schmidt et al., 2017; Wei et al., 2019; Tietz et al., 2013; Kukkk et al., 2021)
Cost	Lower cost, cheaper materials	Higher cost, noble metals	Highest cost	(Kuckshinrichs et al., 2017; Yang et al., 2019; Min et al., 2021)
Offshore suitability	Mature but less flexible	Best match for variable offshore wind	Promising but not proven offshore	(Carmo et al., 2013; Wei et al., 2019; Yu et al., 2018)

Table 1. Simple comparison of AWE, PEM and SOEC for offshore wind-to-hydrogen production.

The comparison shows that PEM electrolysis is the most suitable option for this study because it has fast response, compact design and good suitability for variable offshore

wind. However, PEM still has limitations, including high cost, degradation risk and the need for high purity water.

2.2 Comparative Synthesis and Conflicting Findings

The above three bodies of literature do not lead to one simple ranking of AWE, PEM and SOEC. A recurring weakness is that studies often use different comparison bases. Efficiency is sometimes reported at cell level, stack level or system level, while lifetime data may be measured under steady operation rather than intermittent renewable operation. Cost estimates also vary because studies use different system boundaries. As a result, the apparent performance differences between AWE, PEM and SOEC may partly reflect different assumptions rather than direct technological superiority (Schmidt et al., 2017).

A second limitation is that many catalyst and electrode improvements are tested in small single-cell experiments over short time periods. Although these studies show useful performance gains, they do not always prove that the same improvements can be scaled to durable multi-megawatt electrolyser stacks for offshore operation (Yu et al., 2018).

A third weakness is that many renewable coupling studies treat electrolyzers as isolated units and use idealised power profiles, which poorly represents real offshore wind variability (Lee et al., 2021). PEM is selected as the most defensible modelling basis because it combines dynamic response, compactness, hydrogen purity and offshore operability (Carmo et al., 2013; Wei et al., 2019). However, limited offshore validation, degradation, water purity demand and high cost remain critical uncertainties (DESNZ, 2024; IEA, 2024; ITM Power, 2025).

2.3 Research Gap

Three main gaps are identified from the reviewed literature. First, many studies compare AWE, PEM and SOEC technologies using general performance indicators, but fewer studies assess these technologies against the practical requirements of variable offshore wind operation. This is important because offshore systems require fast response, stable efficiency and good durability under changing power input. PEM electrolysis is often considered suitable for renewable operation, but its cost and degradation under intermittent conditions remain key limitations (Buttler and Spliethoff, 2018; Wei et al., 2019).

Second, existing process studies often treat hydrogen production as a separate unit rather than part of a full offshore power-to-ammonia pathway. This can underestimate the wider energy demand from seawater desalination, water purification, nitrogen supply, compression, ammonia synthesis and liquefaction (Lee et al., 2021; IRENA and AEA, 2022). Therefore, a more integrated modelling boundary is required.

Third, limited published work was found that addresses a modular offshore power-to-ammonia system specifically for the Shetland area and analysing the overall mass balance, total energy balance and techno-economic viability of this facility.

Despite extensive comparison of electrolysis technologies, current literature remains constrained by inconsistent system boundaries and limited validation under offshore intermittency. Therefore, a fully integrated, site-specific modelling approach is required to critically evaluate whether laboratory scale performance claims translate into realistic offshore techno-economic viability.

3. Methodology

3.1 Research Design and System Boundary

This study uses a quantitative process-modelling approach to assess an offshore power-to-ammonia system at the Arven Offshore Wind Site in the East Shetland Basin, Scotland. The system boundary starts from offshore wind power and seawater intake and ends with refrigerated liquid ammonia production for transport. This boundary is selected because the research problem is not only hydrogen production, but the full link between renewable electricity, water treatment, PEM electrolysis, nitrogen supply, ammonia synthesis, storage and techno-economic feasibility.

The 50 MW scale is selected as a modular base case because recent industrial electrolyser developments show that 50 MW systems are being promoted as scalable units for large green hydrogen projects. This makes the model more realistic than a very small laboratory-scale case, while remaining manageable for Aspen Plus simulation and proposal-level techno-economic analysis. The main Aspen Plus model is based on a 50 MW modular offshore production unit. A 50% capacity factor is applied, giving 25 MW of effective available power for PEM electrolysis (ITM Power, 2025; Sunfire, 2026). The full Arven project has a planned capacity of about 2300 MW, but the full-scale plant is not directly modelled because it would require industrial recycle loops, multi-stage compression, heat integration and dynamic control (Ocean Winds, 2025). Therefore, the 50 MW model is used as the base case, while scale-up to 2300 MW is discussed only as an industrial scenario.

Research problem	Related objective	Method used
Variable offshore wind can reduce electrolyser performance	Objective 1	Compare AWE, PEM and SOEC using flexibility, efficiency, durability, cost and suitability for the Arven site.

The full offshore process chain needs to be modelled together	Objective 2	Develop a 50 MW Aspen Plus model including desalinated water supply, PEM hydrogen production, nitrogen supply and ammonia synthesis.
The system mass and energy flows must be clearly quantified	Objective 3	Calculate available wind power, water demand, hydrogen production rate, nitrogen requirement, ammonia production rate and parasitic energy demands.
Hydrogen storage and transport from Shetland are challenging	Objective 4	Assess how hydrogen buffering and conversion to ammonia affect system efficiency, process complexity, safety and offshore logistics.
Economic feasibility of offshore green ammonia is uncertain	Objective 5	Estimate CAPEX, OPEX, LCOA and simple payback period, then test sensitivity to capacity factor, electrolyser energy use, ammonia conversion and production rate.

Table 2. Link between research problems, objectives and methodology.

Research Objectives and Methods

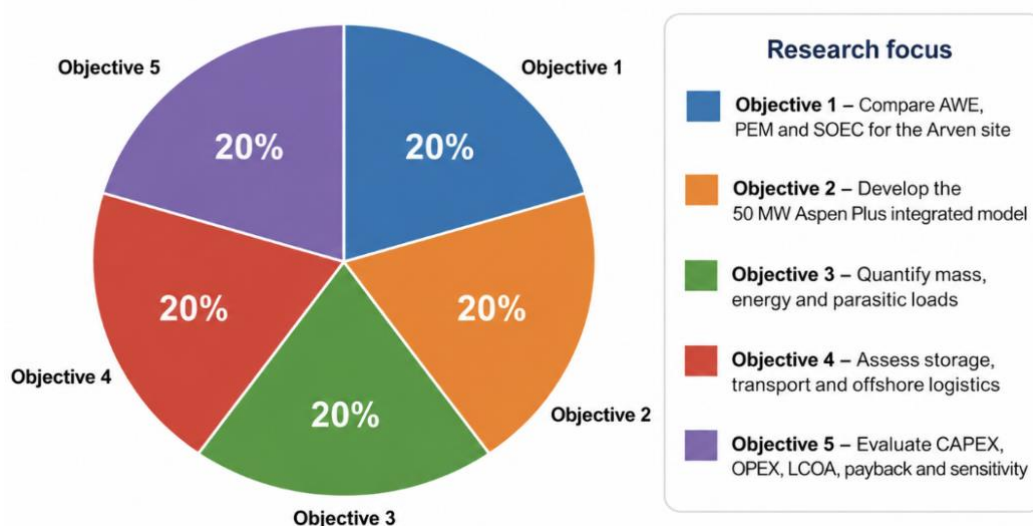


Figure 5. Conceptual overview of research objectives and methodology focus. Equal 20% segments are illustrative only and do not represent workload, time allocation or relative importance.

Figure 5 provides a visual summary of the five research objectives and their related methodology focus. The equal 20% segments are illustrative only and do not represent workload, time allocation or relative research importance.

3.2 Technology Selection

AWE, PEM and SOEC are compared using criteria directly linked to offshore wind operation: response time, efficiency, durability, cost and suitability for remote offshore deployment. PEM electrolysis is selected because it has fast dynamic response, compact design and good suitability for variable renewable electricity (Carmo et al., 2013; Wei et al., 2019). However, this selection is not treated as risk-free. PEM has higher cost, requires high-purity water and may experience membrane or catalyst degradation under fluctuating operation (Schmidt et al., 2017). This critical comparison directly addresses the first objective and justifies why PEM is used in the Aspen Plus model.

3.3 Aspen Plus Process Modelling

The process will be modelled in Aspen Plus using a simplified but technically justified offshore system. Seawater cannot be used directly in PEM electrolysis because salts and impurities can damage the membrane and reduce stack life. Therefore, the model includes a water treatment stage before electrolysis, using desalination and purification to produce deionised water suitable for PEM operation. ELECNRTL will be used for the saline-water section because the model must account for electrolyte behaviour during desalination and purification. This is preferred over a simple non-electrolyte property method because seawater contains dissolved ionic species, and the water-treatment energy demand will be treated as a parasitic load affecting PEM efficiency and total system performance.

Parameter	Value	Unit	Use in model
Installed capacity	50 (ITM Power, 2025; Sunfire, 2026)	MW	Base-case model
Capacity factor	50	%	Available power calculation

Available power	25	MW	Power to PEM
PEM energy use	50	kWh/kg H ₂	Hydrogen production
Water feed	4,500	kg/h	Desalinated water input
H₂ production	500	kg/h	Main PEM output
N₂ feed	2,300	kg/h	Feed to ammonia reactor
PEM temperature	70	°C	Electrolyser condition
PEM pressure	20	bar	Electrolyser condition
NH₃ reactor temperature	450	°C	Reactor condition
NH₃ reactor pressure	150	bar	Reactor condition
NH₃ conversion	0.30	-	Reactor conversion
Actual NH₃ output	20	t/day	TEA production basis

Table 3. Main Aspen Plus design assumptions. The source and calculation basis for key values is provided in Appendix 3.

3.4 Mass and Energy Analysis

The mass and energy balance will be used to check whether the 50 MW Aspen Plus base case is technically consistent. The model uses a 50 MW modular offshore unit with a 50% capacity factor. Therefore, the effective power available for PEM electrolysis is:

$$P_{\text{available}} = P_{\text{installed}} \times \text{CF}$$

$$P_{\text{available}} = 50 \text{ MW} \times 0.50 = 25 \text{ MW}$$

where $P_{\text{available}}$ is the power supplied to the PEM electrolyser, $P_{\text{installed}}$ is the installed wind capacity, and CF is the capacity factor.

Hydrogen production is calculated from the available power and the PEM energy use.

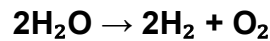
The assumed PEM energy requirement is 50 kWh/kg H_2 :

$$\dot{m}_{H_2} = (P_{\text{available}} \times 1000) / SEC_{\text{PEM}}$$

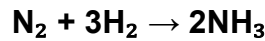
$$\dot{m}_{H_2} = (25 \times 1000) / 50 = 500 \text{ kg/h}$$

where $\dot{m}_{H_2}$ is the hydrogen production rate and SEC_{PEM} is the specific energy consumption of the PEM electrolyser.

The electrolysis reaction used in the model is:



The ammonia synthesis reaction is:



Based on stoichiometry, 1 kg of hydrogen can theoretically produce about 5.67 kg of ammonia therefore:

$$\dot{m}_{NH_3, \text{theo}} = \dot{m}_{H_2} \times 5.67$$

$$\dot{m}_{NH_3, \text{theo}} = 500 \times 5.67 = 2,835 \text{ kg/h}$$

$$\text{Daily theoretical } NH_3 \text{ output} = 2,835 \times 24 = 68,040 \text{ kg/day} \approx 68 \text{ t/day}$$

Ammonia reactor conversion assumption of 0.30 as an optimistic upper-bound case (Rouwenhorst et al., 2021):

$$\dot{m}_{NH_3, \text{actual}} = \dot{m}_{NH_3, \text{theo}} \times X_r$$

$$\text{Actual } NH_3 \text{ output} = \text{Theoretical } NH_3 \text{ output} \times \text{Reactor conversion}$$

$$\text{Actual } NH_3 \text{ output} = 68 \times 0.30 = 20.4 \text{ t/day} \approx 20 \text{ t/day}$$

This 30% value is not treated as a fixed industrial single-pass conversion. Haber-Bosch conversion depends on recycle loops, pressure, temperature and separation; lower conversion and parasitic-load cases will be tested to avoid overestimating ammonia output, net energy and TEA performance (Rouwenhorst et al., 2021).

The net power can be expressed as:

$$P_{\text{net}} = P_{\text{available}} - P_{\text{parasitic}}$$

This step is important because it prevents the system performance from being overestimated and links the Aspen Plus output directly to the techno-economic analysis.

3.5 Techno-Economic Analysis (TEA)

The techno-economic analysis will be based on the same 50 MW Aspen Plus base case. This keeps the process simulation and economic evaluation consistent. The main economic indicators will be capital expenditure, operating expenditure, levelised cost of ammonia and simple payback period.

3.5.1 CAPEX and OPEX Modelling

CAPEX will be estimated for the main process units including water treatment, PEM electrolysis, hydrogen buffering, compression, nitrogen supply, ammonia synthesis, refrigerated ammonia storage and equipment. Because the system is offshore, an offshore location factor will be included to reflect additional marine installation, maintenance, logistics and access costs. A base value of offshore = 1.5 will be used as an indicative proposal assumption, while 1.2-2.5 will be tested in sensitivity analysis because offshore cost premiums depend strongly on site depth, distance from shore, installation method and maintenance access (Myhr et al., 2014).

$$\text{CAPEX}_{\text{total}} = (\sum C_{\text{equip}} \times f_{\text{offshore}}) + C_{\text{install}} + C_{\text{cont.}}$$

where **CAPEX_{total}** is total capital cost, **C_{equip}** is the equipment cost, **f_{offshore}** is the offshore location factor, **C_{install}** is installation cost, and **C_{cont}** is contingency cost.

Operating expenditure will include maintenance, electricity use, PEM stack replacement, water treatment, compression, nitrogen supply and general operation.

$$\text{OPEX}_{\text{annual}} = \text{C}_{\text{maintenance}} + \text{C}_{\text{electricity}} + \text{C}_{\text{stack}} + \text{C}_{\text{water}} + \text{C}_{\text{operation}}$$

This is included because the cost of ammonia depends not only on equipment purchase, but also on long-term operation and maintenance.

3.5.2 Financial Metrics

The **Levelised Cost of Ammonia (LCOA)** and **Simple Payback Period (SPP)** are calculated to determine the breakeven point:

Model outputs will be checked against mass-balance logic and literature-based ranges. Hydrogen production will be compared with the 25 MW power input and 50 kWh/kg H₂ assumption, while ammonia output will be checked against stoichiometry and the 30% reactor conversion. Sensitivity analysis will then test how changes in capacity factor, PEM energy use, ammonia conversion, CAPEX and OPEX affect LCOA and payback period.

Annual ammonia production is calculated from the daily ammonia output:

$$\text{NH}_3_{\text{annual}} = \text{NH}_3_{\text{daily}} \times 365$$

$$\text{NH}_3_{\text{annual}} = 20 \times 365 = 7,300 \text{ t/year}$$

The annualised capital cost is calculated using the capital recovery factor:

$$\text{CRF} = [i \times (1 + i)^n] / [(1 + i)^n - 1]$$

where i is the discount rate and n is the project lifetime.

$$\text{CAPEX}_{\text{ann.}} = \text{CAPEX}_{\text{total}} \times \text{CRF}$$

The levelised cost of ammonia is calculated as:

$$\text{LCOA} = (\text{CAPEX}_{\text{ann.}} + \text{OPEX}_{\text{annual}}) / \text{NH}_{3,\text{annual}}$$

The simple payback period is calculated as:

$$\text{SPP} = \text{CAPEX}_{\text{total}} / (\text{Revenue}_{\text{annual}} - \text{OPEX}_{\text{annual}})$$

where:

$$\text{Annual Net Cash Flow} = \text{Revenue}_{\text{annual}} - \text{OPEX}_{\text{annual}}$$

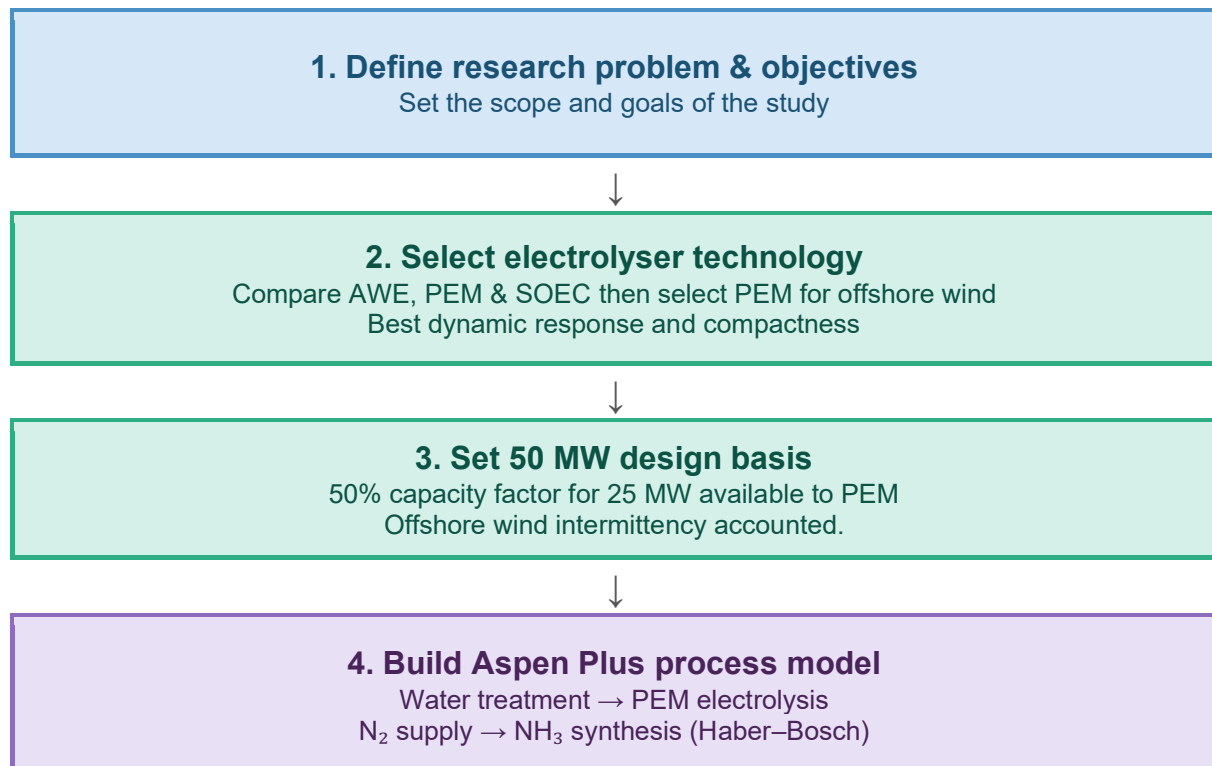
$$\text{Revenue}_{\text{annual}} = \text{NH}_{3,\text{annual}} \times \text{P}_{\text{NH}_3}$$

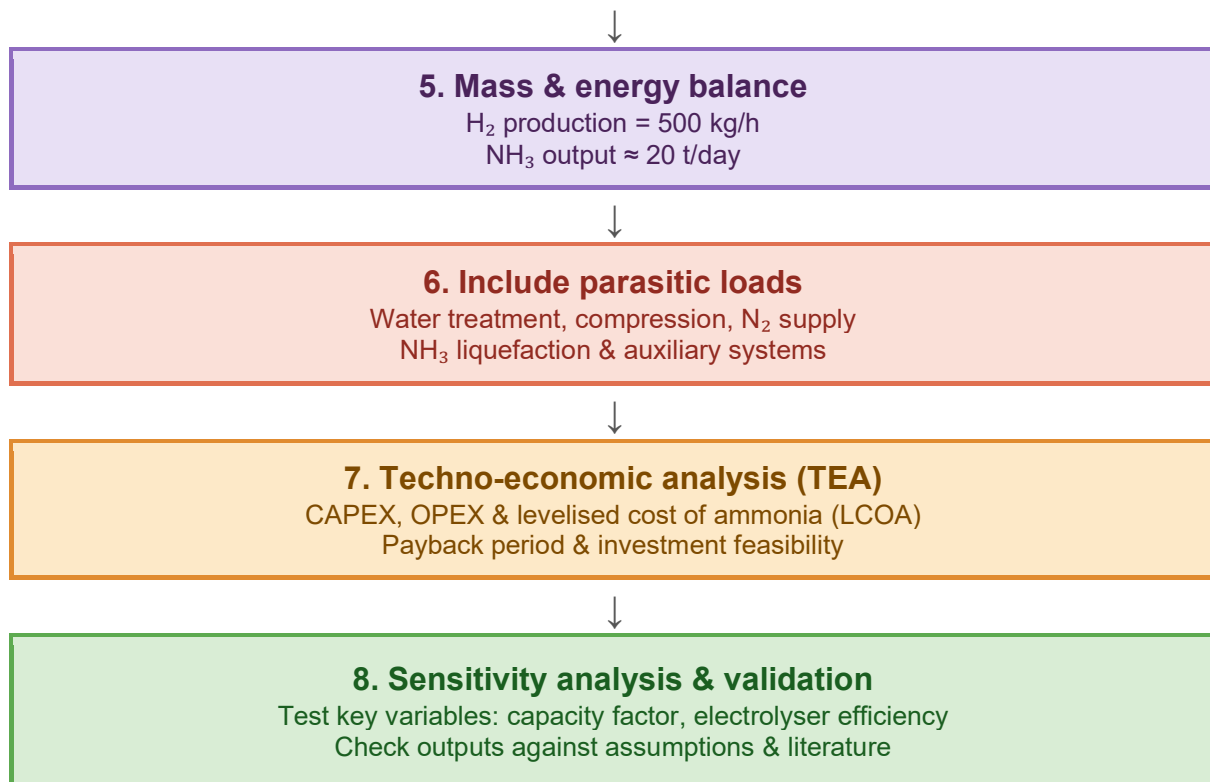
Sensitivity analysis will be used to test how changes in capacity factor, PEM energy use, ammonia conversion, ammonia production rate, CAPEX and OPEX affect LCOA and payback period. The percentage change is calculated as:

$$\Delta\% (\text{Percentage Change}) = (V_{\text{new}} - V_{\text{base}}) / V_{\text{base}} \times 100$$

The final TEA will report a base-case CAPEX, OPEX, LCOA and SPP table, followed by low, base and high sensitivity cases for capacity factor, PEM energy use, ammonia conversion and offshore location factor.

Methodology Flowchart for the 50 MW Offshore Power-to-Ammonia Model:





The flow shows how the methodology moves from problem definition to modelling, TEA and sensitivity testing.

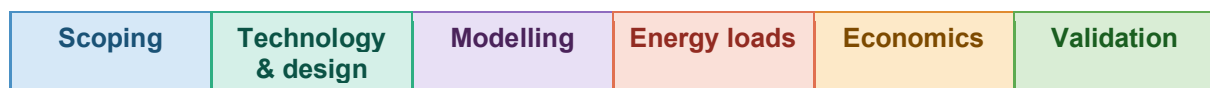


Figure 6. Methodology flowchart for the 50 MW offshore power-to-ammonia model.

Overall, this methodology provides a clear plan for assessing a 50 MW offshore power-to-ammonia system at the Arven site. It links each research objective to a specific method, including technology comparison, Aspen Plus modelling, mass and energy balance, techno-economic analysis and sensitivity testing. Although the model is simplified, it gives a practical early-stage basis for evaluating the technical and economic feasibility of offshore green ammonia production.

4. Expected Contributions

This research will contribute to the assessment of offshore green hydrogen and ammonia production by developing a clear 50 MW modular power-to-ammonia model for the Arven site in Shetland. The study will connect offshore wind power, water treatment, PEM electrolysis, hydrogen buffering, nitrogen supply, ammonia synthesis

and ammonia transport within one system boundary. This will help show how offshore renewable electricity can be converted into a transportable low-carbon fuel.

The project will also provide useful mass balance, energy balance and techno-economic results from Aspen Plus. These results will help evaluate hydrogen production, ammonia output, power demand, capital cost, operating cost and possible economic feasibility. The study is expected to identify the main technical and economic factors that affect the success of offshore green ammonia production, such as capacity factor, electrolyser energy use, ammonia conversion and system cost.

Overall, the research will provide a practical early-stage framework for future offshore hydrogen and ammonia projects. It will not be a full industrial plant design, but it will create a strong basis for future work involving detailed recycle loops, heat integration, dynamic wind modelling and full-scale optimisation. A full life-cycle assessment is outside the scope of this proposal, but the dissertation will identify emissions-related limitations and recommend future LCA work covering offshore electricity supply, water treatment, electrolysis, ammonia synthesis, storage and transport.

5. Ethical Considerations

This project has low risk because the results are derived from simulating processes, using published data and conducting technical analyses. There will be no human participants or data collection by surveys, interviews or the collection of personal data with this project therefore, consent and privacy issues are not expected to arise.

The basic ethical obligation for this project is to accurately and fairly present data. Assumptions, sources and calculations will be clearly referenced so that the results are transparent and not overstated. The Aspen Plus model simulating the ammonia plant is a simplified model and many features of a complete industrial ammonia plant have been left out of the modelling such as recycle loops, detailed heat recovery, and very complex process control.

This project also recognises the issues surrounding the safety and environmental concerns of hydrogen (flammable) and ammonia (toxic). Any future system developed in the real world would need to be designed safely and equipped with risk management

procedures. Even though this project does not chemically handle any material as part of its work and has no real chemical handling risk factors, the final report will also refer to the risks of using hydrogen and ammonia, as well as the project's limitations and responsible conclusions about the proposed system.

Timeline

The Gantt chart was developed in Microsoft Excel to plan the dissertation timeline from February to September 2026. The schedule is organised around project initiation, proposal development, literature review, Aspen Plus modelling, mass and energy analysis, techno-economic assessment, sensitivity analysis, writing, review and final submission. This structure links each project stage to the research objectives and expected outputs.

Task	Start Date	End Date	Feb 2026	Mar 2026	Apr 2026	May 2026	Jun 2026	Jul 2026	Aug 2026	Sep 2026	Notes / Status
PROJECT INITIATION											
Topics Released	03-Feb-26	03-Feb-26	◆								W1/S2
Topic Choices Submission	03-Feb-26	20-Feb-26									Due 4:00 PM
Topic Confirmation	26-Feb-26	26-Feb-26	◆								Due 4:00 PM
SCOPING & PROPOSAL											
Supervisor Meetings & Finalise Scope	04-Mar-26	16-Apr-26									Done by supervisor
Project Proposal Report Submission	04-Mar-26	16-Apr-26									Complete before 16 Apr
INTERIM DELIVERABLES											
Prepare 15-min Presentation Slides	15-Mar-26	07-May-26									Before presentation week
Proposal Feedback	01-Apr-26	24-May-26									Done
Dissertation Proposal Presentation	06-May-26	07-May-26									Before 07/05/2026
Prepare Interim Report	05-May-26	16-Jun-26				◆					Due 4:00 PM
RESEARCH & ANALYSIS											
Detailed Literature Review (AWE vs PEMWE and SOFC)	01-Feb-26	31-May-26									
Offshore Constraints Analysis (Arven Site in Shetland, Scotland)	01-May-26	31-May-26									
Technology Comparison Framework Draft	01-May-26	31-May-26									
Aspen Simulation	01-Jun-26	30-Jun-26									
Synthesis, Conclusions & Research Gaps	01-Jul-26	31-Jul-26									
WRITING & SUBMISSION											
Full Dissertation Draft Writing	01-Jun-26	31-Jul-26									
Supervisor Review & Revisions	01-Jul-26	31-Aug-26									
Final Proofreading & Formatting	01-Aug-26	31-Aug-26									
Report Submission	08-Sep-26	08-Sep-26								◆	Final deadline
Prepared by:											
Syed Youssef Munto	Research	Writing / Presentation	Review	◆ Milestone							

Figure 7. Objective based Gantt chart showing the planned dissertation timeline from February to September 2026.

References

- **Brauns, J. and Thomas, T. (2020)** Alkaline water electrolysis powered by renewable energy: a review [Online]. Available at: <https://doi.org/10.3390/pr8020248> (Accessed: 07 May 2026).
- **Buttler, A. and Spliethoff, H. (2018)** Current status of water electrolysis for energy storage, grid balancing and sector coupling via power-to-gas and power-to-liquids: a review [Online]. Available at: <https://doi.org/10.1016/j.rser.2017.09.003> (Accessed: 14 May 2026).
- **Carmo, M., Fritz, D.L., Mergel, J. and Stolten, D. (2013)** A comprehensive review on PEM water electrolysis [Online]. Available at: <https://doi.org/10.1016/j.ijhydene.2013.01.151> (Accessed: 24 April 2026).
- **Department for Energy Security and Net Zero (DESNZ) (2024)** Hydrogen Strategy Update to the Market: December 2024. London: HM Government. Available at: <https://www.gov.uk/government/publications/uk-hydrogen-strategy> (Accessed: 22 May 2026).
- **David, M., Ocampo-Martinez, C. and Sanchez-Pena, R. (2019)** Advances in alkaline water electrolyzers: a review [Online]. Available at: <https://doi.org/10.1016/j.est.2019.03.001> (Accessed: 24 May 2026).
- **Diogo, M.F.S., Cesar, A.C., Sequeira, J.L. and Figueiredo, L. (2013)** Hydrogen production by alkaline water electrolysis [Online]. Available at: <https://doi.org/10.1590/S0100-40422013000800017> (Accessed: 26 May 2026).
- **Grigoriev, S.A., Fateev, V.N., Bessarabov, D.G. and Millet, P. (2020)** Current status, research trends, and challenges in water electrolysis science and technology [Online]. Available at: <https://doi.org/10.1016/j.ijhydene.2020.03.109> (Accessed: 17 May 2026).
- **HM Government (2021)** Net Zero Strategy: Build Back Greener. London: HM Government. Available at: <https://www.gov.uk/government/publications/net-zero-strategy> (Accessed: 25 March 2026).
- **Hauch, A., Kungas, R., Blennow, P., Hansen, A.B., Hansen, J.B., Mathiesen, B.V. and Mogensen, M.B. (2020)** Recent advances in solid oxide cell technology for electrolysis [Online]. Available at: <https://doi.org/10.1126/science.aba6118> (Accessed: 15 May 2026).

- **ITM Power (2025)** Launch of ALPHA 50 – a new global standard for execution at scale. Available at: <https://itm-power.com/news/launch-of-alpha-50-a-new-global-standard-for-execution-at-scale> (Accessed: 23 May 2026).
- **International Renewable Energy Agency (IRENA) and Ammonia Energy Association (AEA) (2022)** Innovation Outlook: Renewable Ammonia. Abu Dhabi: IRENA. Available at: <https://www.irena.org/publications/2022/May/Innovation-Outlook-Renewable-Ammonia> (Accessed: 28 May 2026).
- **IEA (2024) Global Hydrogen Review 2024.** Paris: International Energy Agency. Available at: <https://www.iea.org/reports/global-hydrogen-review-2024> (Accessed: 08 May 2026).
- **International Energy Agency (IEA) (2021)** Ammonia Technology Roadmap. Paris: IEA. Available at: <https://www.iea.org/reports/ammonia-technology-roadmap> (Accessed: 11 May 2026).
- **Ibrahim, D. (2012)** Green methods for hydrogen production [Online]. Available at: <https://doi.org/10.1016/j.ijhydene.2011.03.173> (Accessed: 03 May 2026).
- **Khan, M.A., Zhao, H., Zou, W., Chen, Z., Cao, W., Fang, J., Xu, J., Zhang, L. and Zhang, J. (2018)** Recent progresses in electrocatalysts for water electrolysis [Online]. Available at: <https://doi.org/10.1007/s41918-018-0014-z> (Accessed: 13 May 2026).
- **Kuckshinrichs, W., Ketelaer, T. and Koj, J.C. (2017)** Economic analysis of improved alkaline water electrolysis [Online]. Available at: <https://doi.org/10.3389/fenrg.2017.00001> (Accessed: 03 May 2026).
- **Kukk, F., Moller, P., Kanarbik, R. and Nurk, G. (2021)** Study of long-term stability of Ni-Zr_{0.92}Y_{0.08}O₂ | Zr_{0.92}Y_{0.08}O₂ | Ce_{0.9}Gd_{0.1}O₂ | Pr_{0.6}Sr_{0.4}CoO₃ at SOFC and SOEC mode [Online]. Available at: <https://doi.org/10.3390/en14040824> (Accessed: 01 April 2026).
- **Lee, B., Lim, D., Lee, H. and Lim, H.K. (2021)** Which water electrolysis technology is appropriate? critical insights of potential water electrolysis for green ammonia production [Online]. Available at: <https://doi.org/10.1016/j.rser.2021.110963> (Accessed: 20 May 2026).
- **Min, X., Shi, Y., Lu, Z., Shen, L., Ogundipe, T.O., Gupta, P., Wang, C., Guo, C., Wang, Z., Tan, H., Mukerjee, S. and Yan, C. (2021)** High performance and cost-effective supported IrO_x catalyst for proton exchange membrane water

electrolysis [Online]. Available at:

<https://doi.org/10.1016/j.electacta.2021.138391> (Accessed: 01 May 2026).

- **Myhr, A., Bjerkseter, C., Ågotnes, A. and Nygaard, T.A. (2014)** 'Levelised cost of energy for offshore floating wind turbines in a life cycle perspective', *Renewable Energy*, 66, pp. 714–728. Available at: <https://doi.org/10.1016/j.renene.2014.01.017> (Accessed: 11 May 2026).
- **Nechache, A. and Hody, S. (2021)** Alternative and innovative solid oxide electrolysis cell materials: a short review [Online]. Available at: <https://doi.org/10.1016/j.rser.2021.111322> (Accessed: 18 May 2026).
- **Nikolaidis, P. and Poullikkas, A. (2017)** A comparative overview of hydrogen production processes [Online]. Available at: <https://doi.org/10.1016/j.rser.2016.09.044> (Accessed: 24 May 2026).
- **Ocean Winds (2025) Arven.** Available at: <https://www.oceanwinds.com/projects/arven/> (Accessed: 13 May 2026).
- **Rouwenhorst, K.H.R., van der Ham, A.G.J. and Lefferts, L. (2021)** 'Beyond Haber-Bosch: The renaissance of the Claude process', *International Journal of Hydrogen Energy*, 46(41), pp. 21566–21579. Available at: <https://doi.org/10.1016/j.ijhydene.2021.04.014> (Accessed: 01 May 2026).
- **Sunfire (2026)** Sunfire launches new 50-megawatt electrolyzer system for the next phase of industrial green hydrogen scale-up. Available at: <https://sunfire.de/en/news/sunfire-launches-new-50-megawatt-electrolyzer-system-for-the-next-phase-of-industrial-green-hydrogen-scale-up/> (Accessed: 11 May 2026).
- **Schmidt, O., Gambhir, A., Staffell, I., Hawkes, A., Nelson, J. and Few, S. (2017)** Future cost and performance of water electrolysis: an expert elicitation study [Online]. Available at: <https://doi.org/10.1016/j.ijhydene.2017.10.045> (Accessed: 15 May 2026).
- **Shiva Kumar, S. and Lim, H. (2022)** An overview of water electrolysis technologies for green hydrogen production [Online]. Available at: <https://doi.org/10.1016/j.egy.2022.10.127> (Accessed: 08 April 2026).
- **Shiva Kumar, S., Ramakrishna, S.U.B., Srinivasulu Reddy, D., Bhagawan, D. and Himabindu, V. (2017)** Synthesis of polysulfone and zirconium oxide coated

asbestos composite separators for alkaline water electrolysis [Online]. Available at: <https://doi.org/10.16966/2473-2575.135> (Accessed: 19 May 2026).

- **Tajuddin, A.A.H., Elumalai, G., Xi, Z., Hu, K., Jeong, S., Nagasawa, K., Fujita, J., Sone, Y. and Ito, Y. (2021)** Corrosion-resistant non-noble metal electrodes for PEM-type water electrolyzer [Online]. Available at: <https://doi.org/10.1016/j.ijhydene.2021.09.116> (Accessed: 07 May 2026).
- **Tietz, F., Sebold, D., Brisse, A. and Schefold, J. (2013)** Degradation phenomena in a solid oxide electrolysis cell after 9000 h of operation [Online]. Available at: <https://doi.org/10.1016/j.jpowsour.2012.09.061> (Accessed: 26 May 2026).
- **Wang, Y., Li, W., Ma, L., Li, W. and Liu, X. (2020)** Degradation of solid oxide electrolysis cells: phenomena, mechanisms, and emerging mitigation strategies – a review [Online]. Available at: <https://doi.org/10.1016/j.jmst.2019.07.026> (Accessed: 19 May 2026).
- **Wei, A., Siebel, A., Bernt, M., Shen, T.H., Tileli, V. and Gasteiger, H.A. (2019)** Impact of intermittent operation on lifetime and performance of a PEM water electrolyzer [Online]. Available at: <https://doi.org/10.1149/2.0421908jes> (Accessed: 12 April 2026).
- **Xie, Z., Yu, S., Yang, G., Li, K., Ding, L., Wang, W., Cullen, D.A., Meyer, H.M., Retterer, S.T., Wu, Z., Sun, J., Gao, P.X. and Zhang, F.Y. (2021)** Ultrathin platinum nanowire based electrodes for high-efficiency hydrogen generation in practical electrolyzer cells [Online]. Available at: <https://doi.org/10.1016/j.cej.2020.128333> (Accessed: 24 May 2026).
- **Yang, G., Yu, S., Mo, J., Kang, Z., Dohrmann, Y., List, F.A., Green, J.B., Babu, S.S. and Zhang, F.Y. (2019)** A novel PEMEC with 3D printed non-conductive bipolar plate for low-cost hydrogen production from water electrolysis [Online]. Available at: <https://doi.org/10.1016/j.enconman.2018.12.046> (Accessed: 02 May 2026).
- **Yu, F., Yu, L., Mishra, I.K., Yu, Y., Ren, Z.F. and Zhou, H.Q. (2018)** Recent developments in earth-abundant and non-noble electrocatalysts for water electrolysis [Online]. Available at: <https://doi.org/10.1016/j.mtphys.2018.11.007> (Accessed: 24 April 2026).

Appendices

Appendix 1. Presents the data used to create Figure 1 in the Introduction on excel. It shows the comparison between the full Arven planned capacity, the 50 MW Aspen Plus model input, and the 25 MW available power after applying a 50% capacity factor.

Category	Value (MW)	% of Arven 2.3 GW	Source / basis
Full Arven planned capacity	2,300	100.00%	Arven official project information / Energy Pathfinder
50 MW Aspen Plus model input	50	2.17%	Uploaded Aspen Plus design assumptions
Available power to PEM (50% CF)	25	1.09%	50 MW × 50% capacity factor
Note: For the final Word proposal, use a logarithmic axis if showing 2300 MW, 50 MW and 25 MW together, because the full Arven project is much larger than the model scale.			
Sources to keep in Excel file			
Arven official project page	https://www.arvenoffshorewind.com/the-project/		
Energy Pathfinder project page	https://energypathfinder.nstauthority.co.uk/projects/2181		
Uploaded design basis	Assumptions		

Appendix 2. Presents the data used for Figure 5. The equal 20% weighting is used only for visual summary and does not represent research importance, workload or time allocation.

Research Objectives and Methods — Arven Offshore Power-to-Ammonia Study		
Objective	Research Focus	Weight (%)
Objective 1	Compare AWE, PEM and SOEC for the Arven site	20%
Objective 2	Develop the 50 MW Aspen Plus integrated model	20%
Objective 3	Quantify mass, energy and parasitic loads	20%
Objective 4	Assess storage, transport and offshore logistics	20%
Objective 5	Evaluate CAPEX, OPEX, LCOA, payback and sensitivity	20%

Appendix 3. Source and calculation basis for Table 3 design assumptions.

Parameter	Source / calculation basis
Installed capacity	Project design assumption, justified by modular 50 MW electrolyser systems (ITM Power, 2025; Sunfire, 2026).
Capacity factor	Project design assumption used to represent offshore wind variability.
Available power	Calculated as $50 \text{ MW} \times 0.50 = 25 \text{ MW}$.
PEM energy use	Proposal-level PEM assumption of 50 kWh/kg H ₂ , used to estimate hydrogen production.
Water feed	Calculated from electrolysis stoichiometry: 1 kg H ₂ requires about 9 kg H ₂ O, so 500 kg/h H ₂ requires about 4,500 kg/h water.
H₂ production	Calculated as $25,000 \text{ kWh/h} \div 50 \text{ kWh/kg H}_2 = 500 \text{ kg/h H}_2$.
N₂ feed	Calculated from Haber-Bosch stoichiometry: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. For 500 kg/h H ₂ , the N ₂ requirement is about 2,300 kg/h.
PEM temperature	Proposal-level design assumption for PEM electrolyser operation, based on typical low-temperature PEM operation (Carmo et al., 2013).
PEM pressure	Proposal-level design assumption for pressurised PEM operation.
NH₃ reactor temperature	Proposal-level Haber-Bosch reactor condition used for early-stage Aspen Plus modelling.
NH₃ reactor pressure	Proposal-level Haber-Bosch reactor condition used for early-stage Aspen Plus modelling.
NH₃ conversion	0.30 used as an optimistic proposal-level upper-bound assumption; lower conversion values will be tested in sensitivity analysis (Rouwenhorst et al., 2021).
Actual NH₃ output	Calculated from 68 t/day theoretical NH ₃ output $\times$ 0.30 reactor conversion = 20.4 t/day, rounded to 20 t/day.